

```
C:\WINDOWS\system32\cmd.exe
Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    IP Address . . . . . : 10.0.0.10
    Subnet Mask . . . . . : 255.0.0.0
    Default Gateway . . . . . : 10.0.0.1

C:\Documents and Settings\user>ping 10.0.0.1

Pinging 10.0.0.1 with 32 bytes of data:

Reply from 10.0.0.1: bytes=32 time=2ms TTL=255
Reply from 10.0.0.1: bytes=32 time=1ms TTL=255
Reply from 10.0.0.1: bytes=32 time=1ms TTL=255
Reply from 10.0.0.1: bytes=32 time=1ms TTL=255

Ping statistics for 10.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\Documents and Settings\user>_

Local Area Connection
LAN or High-Speed Internet

Start | Network Connections | C:\WINDOWS\system32\cmd.exe | 7:00 PM
```

Terminal

R1 x Win-1 x

```
R1#
R1#
R1#
R1#
R1#
R1#
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#int e0/0
R1(config-if)#ip add 10.0.0.1 255.0.0.0
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#int e0/1
R1(config-if)#ip add 20.0.0.1 255.0.0.0
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#exit
R1#
*Aug 22 17:57:46.211: %SYS-5-CONFIG_I: Configured from console by console
R1#sh ip int br

```

Interface	IP-Address	OK?	Method	Status	Protocol
Ethernet0/0	10.0.0.1	YES	manual	up	up
Ethernet0/1	20.0.0.1	YES	manual	up	up
Ethernet0/2	unassigned	YES	TFTP	down	down
Ethernet0/3	unassigned	YES	TFTP	down	down

```
R1#
*Aug 22 18:00:15.937: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R1#
```

```
Terminal
R1 * Win-1 * R2 *
R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#int e0/0
R2(config-if)#ip add 20.0.0.2 255.0.0.0
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#int e0/1
R2(config-if)#ip add 30.0.0.1 255.0.0.0
R2(config-if)#no shut
*Aug 22 18:02:15.226: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#exit
R2#
*Aug 22 18:02:28.149: %SYS-5-CONFIG_I: Configured from console by console
R2#sh ip int br
Interface                IP-Address      OK? Method Status      Protocol
Ethernet0/0              20.0.0.2        YES manual  up          up
Ethernet0/1              30.0.0.1        YES manual  up          up
Ethernet0/2              unassigned      YES TFTP   down        down
Ethernet0/3              unassigned      YES TFTP   down        down
R2#
```

Terminal

Win-1 * R1 * R2 * R3 * Win-2 *

```
10.0.0.1/32 is directly connected, Ethernet0/0
20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
20.0.0.0/8 is directly connected, Ethernet0/1
20.0.0.1/32 is directly connected, Ethernet0/1
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#ip route 30.0.0.0 255.0.0.0 20.0.0.2
R1(config)#ip route 40.0.0.0 255.0.0.0 20.0.0.2
R1(config)#exit
R1#
*Aug 22 18:13:51.553: %SYS-5-CONFIG_I: Configured from console by console
R1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
        a - application route
        + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
10.0.0.0/8 is directly connected, Ethernet0/0
10.0.0.1/32 is directly connected, Ethernet0/0
20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
20.0.0.0/8 is directly connected, Ethernet0/1
20.0.0.1/32 is directly connected, Ethernet0/1
30.0.0.0/8 [1/0] via 20.0.0.2
40.0.0.0/8 [1/0] via 20.0.0.2
R1#
```

```

Terminal
Win-1 * R1 * R2 * R3 * Win-2 *
L    20.0.0.2/32 is directly connected, Ethernet0/0
C    30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
L    30.0.0.0/8 is directly connected, Ethernet0/1
L    30.0.0.1/32 is directly connected, Ethernet0/1
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#ip route 10.0.0.0 255.0.0.0 20.0.0.1
R2(config)#ip route 40.0.0.0 255.0.0.0 30.0.0.2
R2(config)#exit
R2#sh ip route
*Aug 22 18:15:42.230: %SYS-5-CONFIG_I: Configured from console by console
R2#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
        a - application route
        + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S    10.0.0.0/8 [1/0] via 20.0.0.1
C    20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
L    20.0.0.0/8 is directly connected, Ethernet0/0
L    20.0.0.2/32 is directly connected, Ethernet0/0
C    30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    30.0.0.0/8 is directly connected, Ethernet0/1
L    30.0.0.1/32 is directly connected, Ethernet0/1
S    40.0.0.0/8 [1/0] via 30.0.0.2
R2#

```

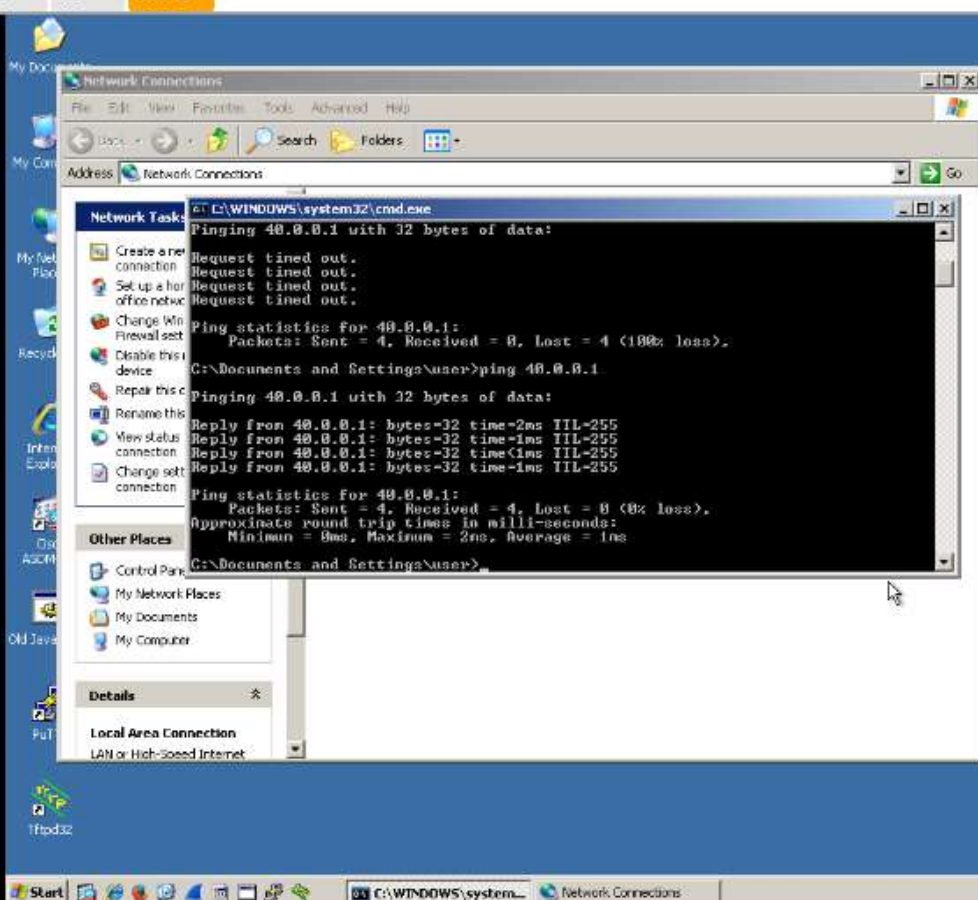
```

Terminal
Win-1 x R1 x R2 x R3 x Win-2 x
L      30.0.0.2/32 is directly connected, Ethernet0/1
C      40.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
L      40.0.0.0/8 is directly connected, Ethernet0/0
L      40.0.0.1/32 is directly connected, Ethernet0/0
R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#ip route 20.0.0.0 255.0.0.0 30.0.0.1
R3(config)#ip route 10.0.0.0 255.0.0.0 30.0.0.1
R3(config)#exit
R3#conf t
*Aug 22 18:17:25.388: %SYS-5-CONFIG_I: Configured from console by console
R3#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S      10.0.0.0/8 [1/0] via 30.0.0.1
S      20.0.0.0/8 [1/0] via 30.0.0.1
C      30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      30.0.0.0/8 is directly connected, Ethernet0/1
L      30.0.0.2/32 is directly connected, Ethernet0/1
C      40.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      40.0.0.0/8 is directly connected, Ethernet0/0
L      40.0.0.1/32 is directly connected, Ethernet0/0
R3#

```



The screenshot shows a Windows XP desktop environment. A command prompt window is open, displaying the results of several ping commands. The first command is `ping 10.0.0.1`, which shows successful results with 0% loss. The second command is `ping 40.0.0.10` from the directory `C:\Documents and Settings\user`, which shows 'Destination host unreachable' for all four attempts. The third command is another `ping 40.0.0.10` from the same directory, which shows successful results with 0% loss. The fourth command is a third `ping 40.0.0.10` from the same directory, which shows successful results with 100% success and 1ms average time. Below the command prompt, a 'Local Area Connection' window is open, showing 'LAN or High-Speed Internet' as the connection type. The desktop background is blue, and the taskbar at the bottom shows the Start button, several application icons, and the system clock showing 7:10.

```

C:\WINDOWS\system32\cmd.exe
My Ping statistics for 10.0.0.1:
  Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
  Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\Documents and Settings\user>ping 40.0.0.10

Pinging 40.0.0.10 with 32 bytes of data:

Reply from 10.0.0.1: Destination host unreachable.
Reply from 10.0.0.1: Destination host unreachable.
Reply from 10.0.0.1: Destination host unreachable.
Reply from 10.0.0.1: Destination host unreachable.

Ping statistics for 40.0.0.10:
  Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
  Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Documents and Settings\user>ping 40.0.0.10

Pinging 40.0.0.10 with 32 bytes of data:

Reply from 40.0.0.10: bytes=32 time=3ms TTL=125
Reply from 40.0.0.10: bytes=32 time=1ms TTL=125
Reply from 40.0.0.10: bytes=32 time=1ms TTL=125
Reply from 40.0.0.10: bytes=32 time=2ms TTL=125

Ping statistics for 40.0.0.10:
  Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
  Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 3ms, Average = 1ms

C:\Documents and Settings\user>
  
```

Local Area Connection
LAN or High-Speed Internet

Start Network Connections C:\WINDOWS\system32\cmd.exe 7:10

Terminal

R1 x Win-1 x

```
R1#
R1#
R1#
R1#
R1#
R1#
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#int e0/0
R1(config-if)#ip add 10.0.0.1 255.0.0.0
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#int e0/1
R1(config-if)#ip add 20.0.0.1 255.0.0.0
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#exit
R1#
*Aug 22 17:57:46.211: %SYS-5-CONFIG_I: Configured from console by console
R1#sh ip int br

```

Interface	IP-Address	OK?	Method	Status	Protocol
Ethernet0/0	10.0.0.1	YES	manual	up	up
Ethernet0/1	20.0.0.1	YES	manual	up	up
Ethernet0/2	unassigned	YES	TFTP	down	down
Ethernet0/3	unassigned	YES	TFTP	down	down

```
R1#
*Aug 22 18:00:15.937: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R1#
```

Terminal

R1 * Win-1 *

```
C:\WINDOWS\system32\cmd.exe
Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    IP Address . . . . . : 10.0.0.10
    Subnet Mask . . . . . : 255.0.0.0
    Default Gateway . . . . . : 10.0.0.1

C:\Documents and Settings\user>ping 10.0.0.1

Pinging 10.0.0.1 with 32 bytes of data:

Reply from 10.0.0.1: bytes=32 time=2ms TTL=255
Reply from 10.0.0.1: bytes=32 time=1ms TTL=255
Reply from 10.0.0.1: bytes=32 time=1ms TTL=255
Reply from 10.0.0.1: bytes=32 time=1ms TTL=255

Ping statistics for 10.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\Documents and Settings\user>_
```

Local Area Connection
LAN or High-Speed Internet

Start Network Connections C:\WINDOWS\system32\cmd.exe 7:00 PM